Smart Pothole Detection and Monitoring System

ESP32-Based Pothole Detection and Maintenance Alert System

Introduction

- Potholes are a major concern for road safety and vehicle maintenance.
- Traditional road inspections require manual effort and time.
- An automated monitoring system can help identify road damage efficiently.
- Our project uses ESP32 and multiple sensors to detect potholes and monitor road conditions.
- Sensor readings are transmitted wirelessly to a web-based dashboard.



Problem Statement

01

Manual pothole identification is time-consuming.

02

Delayed detection and maintenance increase road safety risks.

03

Water-filled potholes are difficult to identify visually.

04

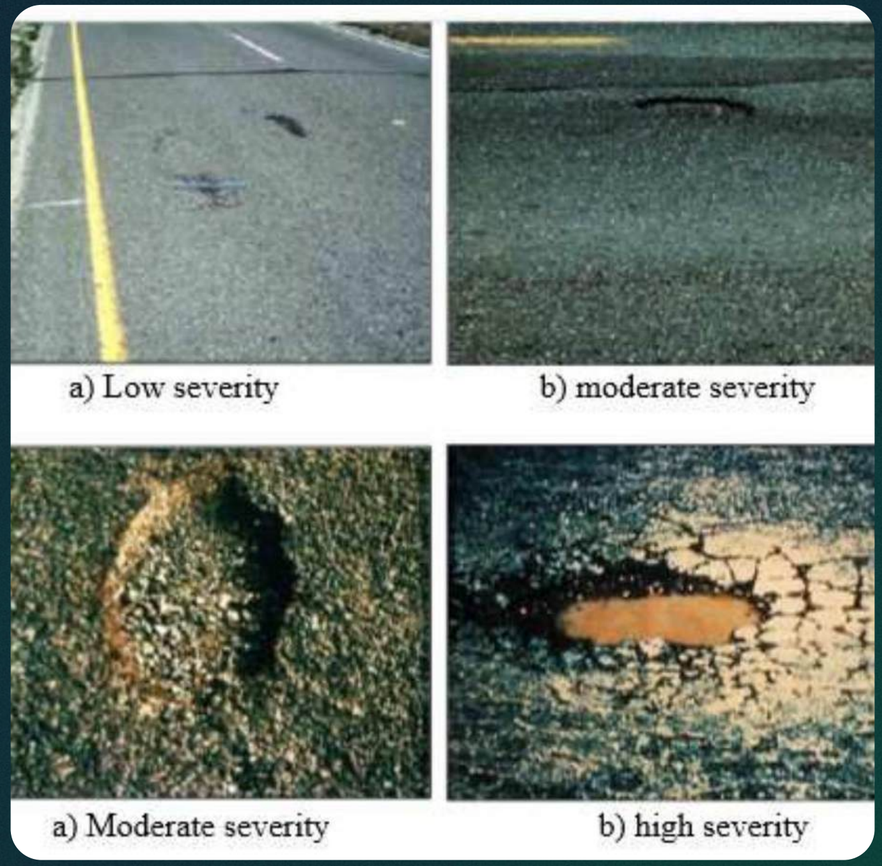
Lack of continuous monitoring makes maintenance planning challenging.

Our Solution:

Develop an IoT-based system that detects potholes, estimates their severity, identifies water presence, and supports maintenance tracking.

Objectives

01	Detect potholes using an ultrasonic sensor.
02	Estimate pothole depth and classify severity.
03	Identify water presence using a moisture sensor.
04	Estimate vehicle speed using a magnetic sensor.
05	Location information through GPS.
06	Sensor readings on a web dashboard.
07	Maintenance alerts and track repair status.



Hardware Components

01

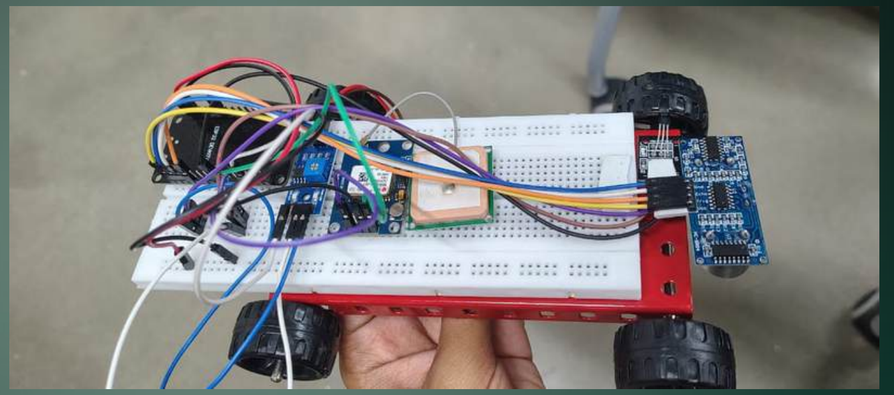
ESP32

Main microcontroller and Wi-Fi communication

02

Ultrasonic Sensor

Measures distance to the road surface



03

Magnetic Sensor

Counts wheel rotations

04

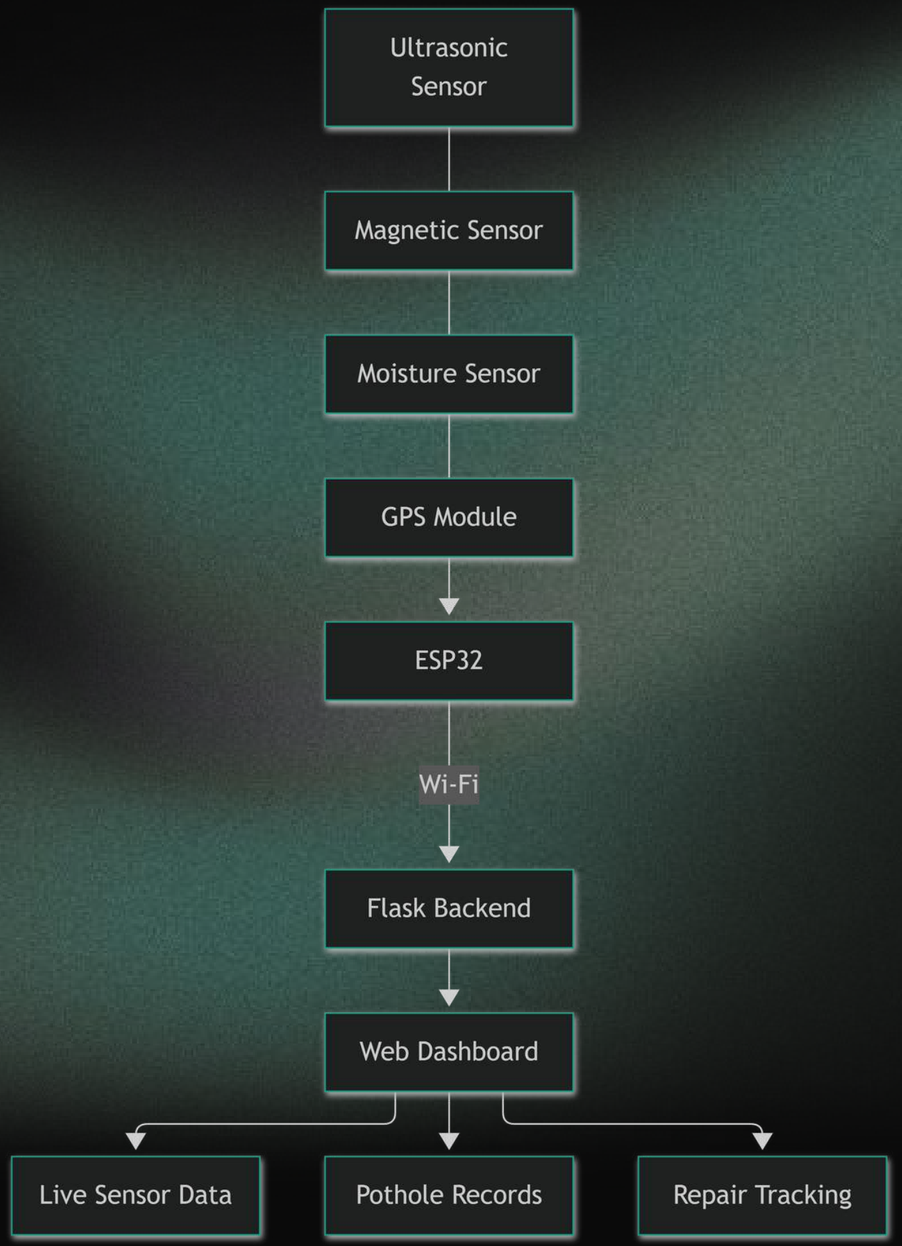
Moisture Sensor

Detects water presence

05

GPS Module

Provides geographical coordinates



System Architecture

Sensors collect road-condition data, which is processed by the ESP32 and transmitted to the Flask backend through Wi-Fi. The website displays the readings and supports maintenance monitoring.

Working Principle

01

Road Surface Monitoring

02

Pothole Detection

03

Additional Monitoring

04

Data Transmission

05

Web Monitoring

Software Implementation

Technology

Arduino IDE: ESP32 programming and sensor interfacing.
Embedded C/C++: Sensor processing and pothole detection.
Python Flask: Backend server and API communication.
HTML: Website structure.
CSS: Dashboard design.
JavaScript: Live sensor monitoring and dashboard updates.

Communication

ESP32 sends sensor data to Flask through HTTP POST requests. The website retrieves the latest readings through HTTP GET requests.

Results and Current Implementation

- Successful ESP32 sensor interfacing.
- Ultrasonic distance and estimated depth measurement.
- Magnetic sensor-based wheel rotation counting.
- Moisture detection.
- GPS integration with simulated-location fallback.
- Successful ESP32-to-Flask communication.
- Live sensor readings displayed on the website.
- Automatic pothole ticket generation.
- Dashboard statistics based on recorded potholes.
- Maintenance status updates.
- Calibration of magnetic sensing and severity thresholds.



Conclusion

- Developed an ESP32-based prototype for monitoring road conditions.
- Integrated ultrasonic, magnetic, moisture, and GPS sensors.
- Established wireless communication between the ESP32 and a web dashboard.
- Demonstrated live sensor monitoring and pothole depth estimation.

Future Scope

- Improve pothole detection accuracy through sensor calibration.
- Implement reliable speed-breaker detection.
- Add permanent database storage for pothole records.
- Improve GPS-based location tracking.
- Develop a more robust maintenance management system.

Thank You

FARAH MANIKINDI – 245859016
SHARADHICP – 245859018
BANI V – 245859022
HARSHITHA GR – 245859072